



Pramaan

Proving a document is real — in seconds.

An AI document-authenticity and fraud-detection system for government schemes, built so that its conclusions can be defended: mathematics decides what mathematics can decide, and the system says plainly what it could not verify.

12

forensic modules per document

40/40

checksum validator tests passing

0

fabricated external lookups

Submitted for hackathon evaluation · Repository: github.com/Yashraj00700/pramaan

A real person, a real scheme, a fake piece of paper

Most welfare fraud in India is not a stolen identity. It is a real, unique person — correctly Aadhaar-authenticated — submitting a document that says something untrue about them.

India already solved a large part of the identity problem. Direct Benefit Transfer removed ghost and duplicate beneficiaries at enormous scale. But Aadhaar proves *who you are*. It does not prove that the income certificate in your file says your real income, that the caste certificate was issued by the officer whose seal it carries, or that the bank guarantee backing a tender is real.

That second gap is still checked the way it was checked decades ago: a clerk looks at the paper. One officer, hundreds of applications, no forensic tooling, no time. A competent forgery passes. An honest applicant with an unusual-looking document sometimes gets rejected. Both failures are invisible, because nobody records them.

The specific thing we set out to build: put a forensic examiner behind every counter — one that reads the document, checks the arithmetic, validates the identifiers against their real checksum rules, argues with itself about the evidence, and then tells the officer exactly what it found and exactly what it could not confirm.

Why Madhya Pradesh is the sharpest case

MP carries a documented institutional memory of exactly this failure mode. The Vyapam scandal turned document and examination fraud into a decade-long investigation. More recently, a ₹202 crore fake bank-guarantee case in the MP Jal Nigam pipeline showed the same weakness on the procurement side: forged instruments accepted because nobody could verify them at the counter.

Leakage is measured, acknowledged, and funded against

₹3.48 lakh crore cumulative DBT savings from removing leakage (Union IT Minister)	5.2 crore fake ration cards purged during that exercise
830 of 1,572 reviewed minority institutions found fake or inactive (₹144.83 cr)	463 of 474 claimed beneficiaries fake in one Kashmir scheme audit

We are deliberately not overclaiming this number. The bulk of DBT's savings came from Aadhaar-based *identity* deduplication — removing ghost and duplicate people — not from document-level forensics. Pramaan targets the complementary failure: the document is fabricated or altered even though the applicant is a real, unique person. We cite DBT to establish that leakage at this scale is real and government-acknowledged, not to claim Pramaan would capture the same ₹3.48 lakh crore.

The document-fraud pattern, case by case

CASE	SCALE	DOCUMENT THAT FAILED
MP Jal Nigam procurement	₹202 crore	Forged bank guarantees accepted as security
Kolkata-syndicate guarantee racket	₹183 crore	Fake bank-guarantee instruments
Himachal scholarship scam	₹181 crore (2013–17)	Fabricated enrolment / institution records
Minority scholarship audit	₹144.83 crore	Fake institutions and beneficiary documents
Ahmedabad GST ITC scam (Dec 2025)	₹1,500 crore	Fake invoices

What we can honestly say: no consolidated public figure exists for "₹X crore lost to fake documents in MP welfare schemes per year" — we looked, and we will not invent one. What the record does establish is that (1) document and eligibility fraud is large enough that the union government funded a multi-lakh-crore programme against one slice of it, (2) MP has open, acknowledged fraud in exactly the pipelines we target, and (3) scholarship fraud in the 25–30% range is documented at national scale.

HOW IT WORKS

Deterministic facts first, AI reasoning second

The order matters. Anything a computer can prove, a computer proves — and the AI is not allowed to argue with it.



The part most tools get wrong

It is trivial to ask a language model "is this document fake?" and print whatever it says. It will confidently report WHOIS records it never fetched and sanctions hits it never checked. That output is worthless to an officer who may have to defend a rejection.

Our rule: the model may only reason from (a) what is visibly on the document, (b) facts computed in code, and (c) searches it actually performed. Anything else is written into a `externalChecksNeeded` list — flagged for a human, never invented.

Two arguments, one adjudicator, binding evidence

A single AI pass tends to either cry fraud at every compression artifact, or wave everything through. We made it argue with itself instead.

Prosecution

Builds the strongest honest case that the document is forged, citing only observable evidence: baseline drift on a re-typed figure, a serial that does not match the issuer's format, an income that contradicts the balance shown.

Defence

Gives the innocent explanation for each indicator — scanner artifacts, JPEG recompression, a legitimate re-save, a template revision — and points out where the prosecution overreaches.

The judge weighs both against the code-computed facts, which are *binding*. A failed checksum cannot be argued away. An OFAC list that failed to download cannot be spun into "clean". The ruling records what was **decisive** and, critically, what it **dismissed** — that dismissal list is the false-positive guard, and it is shown to the officer.

Why this matters for a government deployment

A rejection that cannot be explained is a rejection that gets challenged. Every Pramaan verdict ships with the reasoning chain, the evidence quoted from the document, the arguments that were rejected and why, and an explicit list of what still requires verification at source.

What we actually tested, and what we refuse to claim

Verified

CLAIM	EVIDENCE
Identifier validation is real mathematics	40/40 unit tests pass — Aadhaar (Verhoeff), PAN structure + entity code, GSTIN mod-36, IFSC, IBAN mod-97
Forensic pipeline runs without an API key	npm run selftest → 5/5 pass, no key required
ELA responds to real pixel edits	mean error moved 0.196 → 0.264 when a splice was introduced
Watchlist screening is live, not mocked	screened against the real OFAC SDN export — 19,393 records parsed

Honest limitations

Error-Level Analysis is weak on flat, vector-rendered documents. ELA was designed for photographs. On a crisp certificate, ordinary body text produces as much error as a spliced region. On our own tampered specimen the edited serial rose from ≈ 1.7 to ≈ 4.1 but did not cleanly separate from normal text. We therefore treat ELA as one weak indicative signal, never as a verdict — and we say so in the report rather than letting it look decisive.

- **No signature or seal verification yet.** That needs a vision model behind a separate service; the public reference implementations are `license: null` or AGPL and cannot be used.
- **No source-of-truth verification yet.** Until DigiLocker / e-District integration, Pramaan can say a certificate is internally consistent and structurally valid — not that the issuing office actually issued it.
- **Confidence is self-reported** by the model, not calibrated against a labelled dataset.

Every one of these appears in the product's own reports, not just this document.

What we can connect to, and what that costs

SOURCE	WHAT IT PROVES	REALITY
Aadhaar offline QR / XML	Signed identity data, verified against UIDAI's public key	Buildable now fully offline, no API
OFAC / UN / EU lists	Watchlist screening	Live today free official downloads
ICAO 9303 MRZ	Passport/ID check digits	Pure arithmetic no ML needed
PAN / GSTIN / MCA / IEC	Registry existence and status	Paid aggregator self-serve, same-day
DigiLocker / e-District	The real proof — did this office issue this certificate?	Needs partnership weeks, not hours

The honest sequencing: the checks that need no permission ship first and already carry real weight. DigiLocker is the endgame — it converts Pramaan from "this document is internally sound" to "this document was issued by that office" — and it needs a government sponsor, which is precisely what a department pilot would unlock.

Where it fits operationally

- **Counter-level triage:** the clerk uploads, gets a verdict and a reason, and escalates only what is flagged.
- **Audit trail:** every verdict stores its evidence, so a rejection can be defended and a mistake can be reviewed.
- **Officer stays in charge:** Pramaan recommends; it does not auto-reject. Fraud decisions remain human.

Roadmap, in order of honest value

NEXT	WHY IT EARNS ITS PLACE
ICAO 9303 MRZ check digits	Pure arithmetic on data the model already transcribes — provable ID validation for passports and ID cards, zero new dependencies
PDF xref / incremental-update analysis	Detects whether page content was modified in a later save — catches PDF-certificate forgery far better than metadata alone
JPEG quantisation-table + ghost analysis	The technique that actually works where ELA is weak; buildable in Node
Aadhaar offline-QR signature verification	Cryptographic proof of identity data, entirely offline
DigiLocker / e-District issuer verification	Turns "consistent" into "issued" — the decisive check
Signature & seal verification service	Requires a separate Python/ONNX service; licensing rules out the public SDKs

The ask

A pilot on one scheme, in one department, with real submitted documents and an officer in the loop. That produces the two things we cannot generate ourselves: a labelled dataset to calibrate confidence against, and the sponsorship needed for DigiLocker issuer verification.

What we would measure in a pilot: fraudulent documents caught that manual review missed; false positives (honest applicants wrongly flagged — the number most systems hide); officer time per application; and turnaround from submission to decision.

Sources: Akashvani/News on Air and DBT Bharat (DBT savings and purge figures); Deccan Herald (Himachal scholarship); Angel One / The420.in (fake bank-guarantee cases and Vyapam sentencing); DeshGujarat (Ahmedabad GST ITC). Full citations with URLs in the repository under docs/ .